

1. System Scope

1.1 In Scope

- I. **Geographic Coverage:** Melaka only (prototype phase); expansion to other states is a future consideration, not part of this release.
- II. **Platform:** Progressive Web App (PWA)
- III. **Functional Coverage:** All six core modules — User Management, Attraction Explorer, Review & Community, Exploration Map, Social Profile, Personal Collection
- IV. **Target Users:** Local and international tourists
- V. **Offline Capability (Limited):** App launch, attraction list/detail (previously viewed), cached static images, cached search results, and cached map regions remain accessible offline via PWA Service Worker caching.

1.2 Out of Scope

- I. Regions outside Melaka
- II. Booking / payment functionality
- III. Real-time chat between users
- IV. Admin backend / content moderation panel (no module defines admin-level functions)
- V. Multi-language support
- VI. Native iOS / Android applications
- VII. Full offline write capability — Social Login, Review submission, Social Profile actions, and Collection sync all require an active network connection (see NFR-08 for the complete offline-support matrix)

2. Actor Definitions

Actor	Definition	Applies to
Guest	An unauthenticated visitor who has not registered or logged in. Can browse public content but cannot perform actions tied to a personal account.	Attraction Explorer (search/view only)
Registered User	A user who has successfully logged in through a supported social login provider and has access to the full personal, review, social exploration, and collection functionalities of the system.	All modules

3. Functional Requirements List

FR-ID	Module	Requirement Description
FR-001	User Management	The system shall allow users to log in using supported social login accounts.
FR-002	User Management	The system shall allow logged-in users to log out securely.
FR-003	User Management	The system shall allow users to edit their personal profile information.
FR-004	User Management	The system shall allow users to upload or update their profile picture.
FR-005	Attraction Explorer	The system shall display a default list of Melaka attractions and allow Guest and Registered Users to search the list by attraction name, location, or category on the same Attractions page.
FR-006	Attraction Explorer	The system shall allow Guest and Registered Users to filter attractions by category, location area, and minimum rating, either independently or together with a search keyword. The system shall show the current result count and applied criteria, allow users to clear or reset them, and provide recovery options on the same page when no attractions match.
FR-007	Attraction Explorer	The system shall display the selected attraction's name, description, category, address, and rating.
FR-008	Attraction Explorer	The system shall display attraction photo previews, allow an available photo to be enlarged with previous, next, and close controls, and provide a non-blocking no-photo fallback when no photos are available.
FR-009	Attraction Explorer	The system shall optionally display the selected attraction's location on an interactive map using one marker with the attraction name and address and basic zoom and pan controls. If the map cannot be loaded, the system shall display the address in text form. This function shall not provide current-location tracking, nearby search, directions, route navigation, travel time, transport mode, or turn-by-turn guidance.

FR-010	Review & Community	The system shall allow users to create a review for an attraction.
FR-011	Review & Community	The system shall allow users to give a rating when submitting a review.
FR-012	Review & Community	The system shall allow users to upload photos together with their review.
FR-013	Review & Community	The system shall allow users to edit their own reviews.
FR-014	Review & Community	The system shall allow users to delete their own reviews.
FR-015	Review & Community	The system shall allow users to view reviews posted under each attraction.
FR-016	Exploration Map	The system shall display a personal exploration map for each user.
FR-017	Exploration Map	The system shall automatically highlight attractions that the user has visited.
FR-018	Exploration Map	The system shall allow users to view the list of attractions they have visited.
FR-019	Exploration Map	The system shall track the user's exploration progress based on visited attractions.
FR-020	Social Profile	The system shall allow users to view other users' public profiles.
FR-021	Social Profile	The system shall allow users to view reviews written by other users from their profile page.
FR-022	Social Profile	The system shall allow users to view other users' exploration maps.
FR-023	Social Profile	The system shall allow users to compare exploration progress with other users.
FR-024	Social Profile	The system shall allow users to compare visited attractions with other users.

FR-025	Personal Collection	The system shall allow users to save attractions into their wishlist.
FR-026	Personal Collection	The system shall allow users to manage their favourite attractions.
FR-027	Personal Collection	The system shall allow users to view all reviews they have created.
FR-028	Personal Collection	The system shall allow users to view all photos they have uploaded.
FR-029	Personal Collection	The system shall allow users to view their personal travel history.

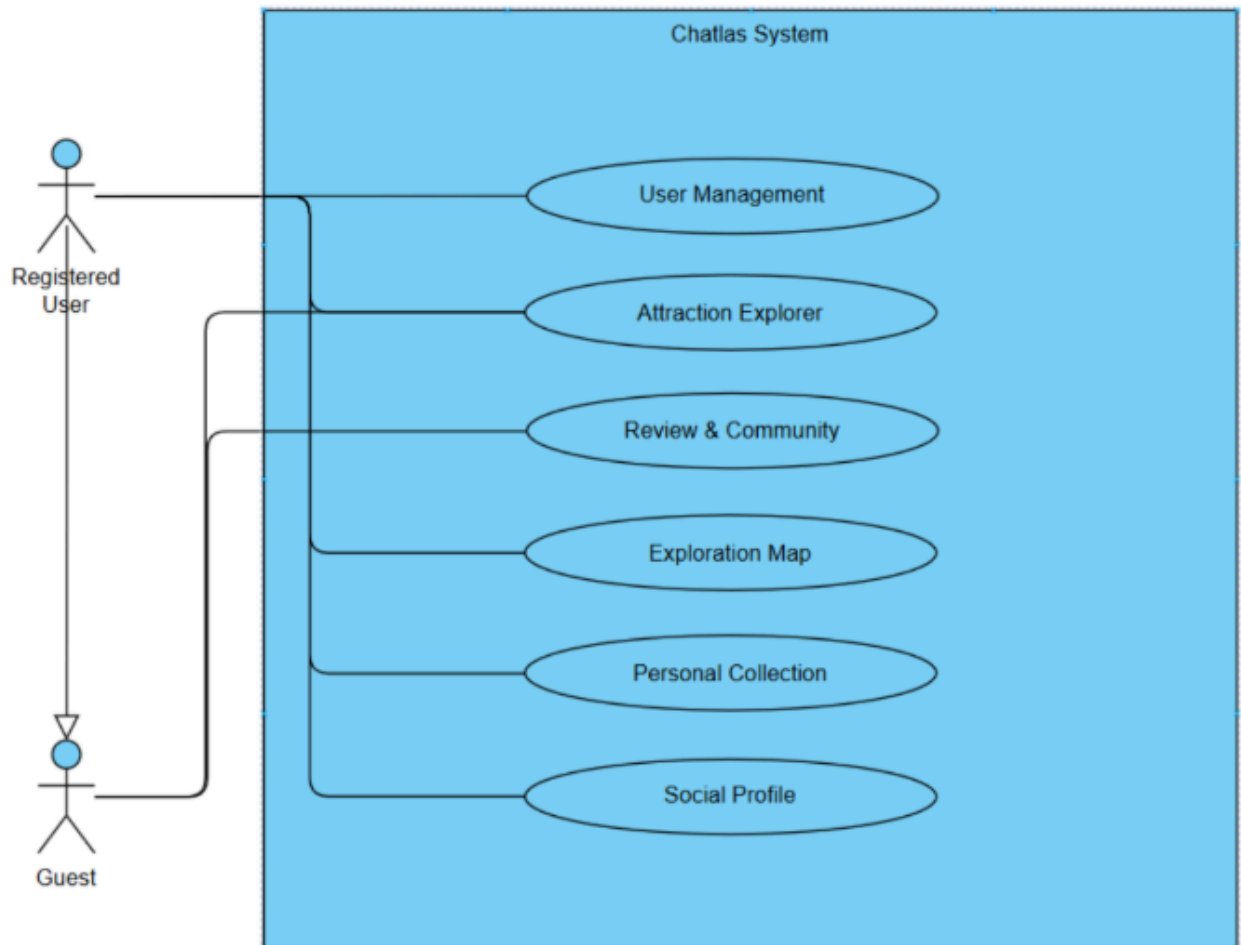
4. Non-Functional Requirements List

NFR-ID	NFR-01
Quality	Performance (Stress Testing)
Description	The system must deliver rapid map rendering and fluid user experience across key discovery modules under varying tourist network conditions.
Measurable Target	a) Over a standard 4G/5G connection, the interactive map layout, custom markers, and Google Places API details for Melaka attractions must fully render within 2.5 seconds. b) When operating offline, previously viewed static assets and cached map regions managed by the PWA Service Worker must load within 0.5 seconds.
Verification / Testing Method	Tool: Google Chrome DevTools (Network & Performance), Lighthouse. Method: Execute the loading test 10 times under 4G/5G simulation and Offline Mode. Pass Criteria: At least 9 out of 10 tests meet the required loading time.

NFR-ID	NFR-02
Quality	Availability
Description	The platform must remain accessible during normal active usage, testing and demonstration periods while providing limited offline access through PWA

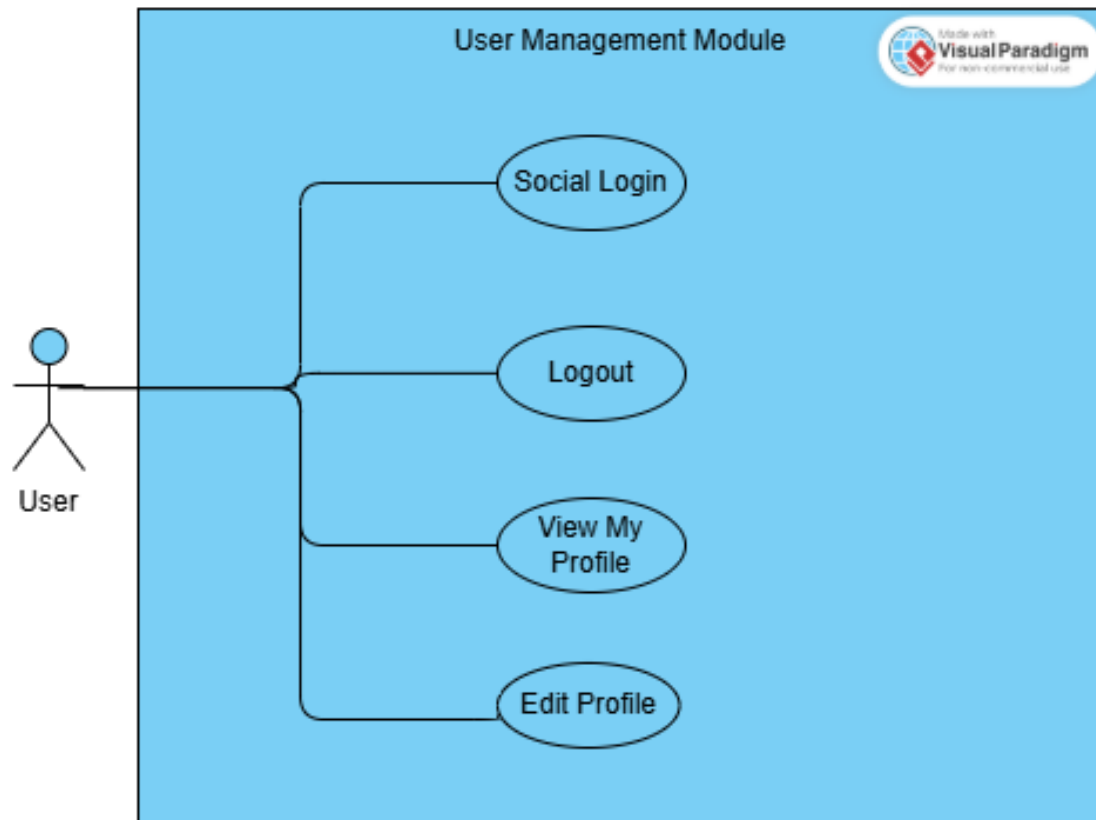
	Service Worker caching when internet connectivity or backend service availability is temporarily interrupted.
Measurable Target	a) During a continuous 30-minute active testing or demonstration session, the core online functionalities must remain accessible without unexpected application crashes after the backend service is active. b) If the backend is temporarily inactive due to free-tier hosting limitations, the system must recover automatically without data loss once the backend service becomes available again. c) During offline testing, the PWA must successfully launch and display previously cached application assets, attraction lists, previously viewed attraction details, static images, cached search results, and cached map regions in 10 out of 10 test attempts after the required resources have been cached. d) Online-only or write operations, including Social Login, Review submission, Social Profile actions, and Collection synchronisation, must display a clear offline or service-unavailable message instead of showing a blank screen or causing the application to crash.
Verification / Testing Method	Tool: Chrome DevTools (Offline Mode), Browser Console, Backend Logs. Method: Run a continuous 30-minute testing session, disconnect the network using Offline Mode, then reconnect. Pass Criteria: No application crashes, cached content loads successfully, and online-only features display appropriate error messages.

5. System-Level Use Case Diagram



6. User Management Module - Use Case Diagram & Description (Tan Yi Jia)

6.1 Use Case Diagram



6.2 Use Case Description

UC-UM-01: Social Login

Element	Description
Use Case ID	UC-UM-01
Use Case Name	Social Login
Primary Actor	User
Description	Allows a user to log in or register using their Google account. If the user is new, the system automatically creates an account. If the user already exists, the system logs them in directly.
Precondition	The user is not currently logged in.
Postcondition	The user is successfully authenticated and logged into the system. If the user is new, a new account is created.

Main Flow	
User	System
1. Open the login page.	
	2. Displays the "Sign in with Google" button.
3. Click the "Sign in with Google" button.	
	4. Redirects to Google for authentication.
	5. Retrieves the user's profile information (name, email) from Google.

	6. Checks if the email is already registered in the database.
	7a. If the email exists: Logs the user in.
	7b. If the email does not exist: Creates a new account and logs the user in.
	8. Creates a user session.
	9. Redirects to the homepage.

Alternative Flow	
Alternative Flow	Description
A1: User cancels Google authentication	The user closes the Google authentication window. The system returns to the login page without logging the user in.
A2: Google authentication fails	The system fails to retrieve user information from Google. The system displays an error message: "Google authentication failed. Please try again."
A3: Email retrieval failed	Google authentication succeeds but the system cannot retrieve the user's email. The system displays an error message and asks the user to try again.

UC-UM-02: Logout

Element	Description
Use Case ID	UC-UM-02
Use Case Name	Logout
Primary Actor	User
Description	Allows a logged-in user to end their session and log out of the system.
Precondition	The user is currently logged in.
Postcondition	The user session is destroyed and the user is redirected to the login page.

Main Flow	
User	System
1. Click the logout button.	
	2. Destroys the user session.
	3. Redirects to the login page with a message: "You have been logged out successfully."

Alternative Flow	
Alternative Flow	Description
A1: Session already expired	1. The user clicks logout but the session has already expired. 2. The system clears any remaining session data and redirects to the login page.

UC-UM-03: View My Profile

Element	Description
Use Case ID	UC-UM-03
Use Case Name	View My Profile
Primary Actor	User
Description	Allows a logged-in user to view their own profile information, including their name, email, and profile picture.
Precondition	The user is currently logged in.
Postcondition	The user's profile information is retrieved from the database and displayed on the screen.

Main Flow	
User	System
1. Navigates to the profile page.	
	2. Retrieves the user's profile information from the database.
	3. Displays the user's profile information (name, email, profile picture).
4. View their profile information.	

Alternative Flow	
Alternative Flow	Description
A1: Profile data cannot be loaded	1. The system fails to retrieve the user's profile data due to network or data issues. 2. The system displays an error message: "Unable to load profile. Please refresh the page."
A2: User session expired	1. The system detects that the user session has expired. 2. The system redirects the user to the login page.

UC-UM-04: Edit Profile

Element	Description
Use Case ID	UC-UM-04
Use Case Name	Edit Profile
Primary Actor	User
Description	Allows a logged-in user to update their personal information. During this process, the user may also upload or update their profile picture.
Precondition	The user is currently logged in.
Postcondition	The user's profile information is successfully updated and saved in the database. If a new profile picture was uploaded, it is also updated.

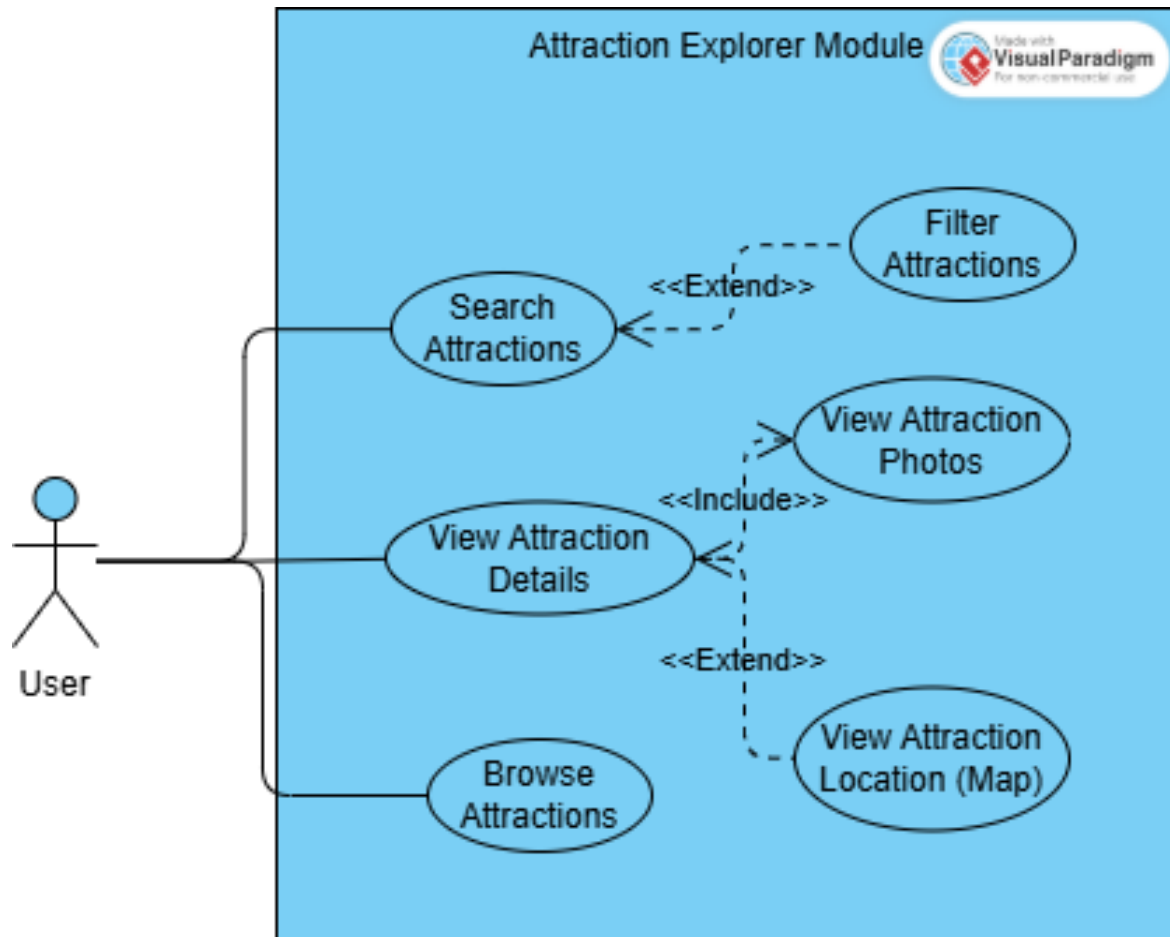
Main Flow	
User	System
1. Navigates to the profile page.	
	2. Displays the user's current profile information.
3. Click the "Edit Profile" button.	
	4. Displays an editable form with the user's current information.
5. Modifies the information (e.g., name, bio).	
6. Optionally uploads a new profile picture.	

	7. Validates the uploaded image format and size (if a photo was uploaded).
	8. Uploads and saves the image to the database (if a photo was uploaded).
9. Submit the form.	
	10. Validates all input information.
	11. Updates the user's profile in the database.
	12. Displays a success message and refreshes the profile page.

Alternative Flow	
Alternative Flow	Description
A1: Unsupported file format	1. The user selects an unsupported file format (e.g., BMP, WEBP). 2. The system displays a message: "Unsupported file format. Please upload a JPG or PNG." 3. The user must select a valid file.
A2: Invalid input format	1. The system displays an error message. 2. The user must correct the input and resubmit.
A3: File too large	1. The user selects a file that exceeds the size limit (e.g., >2MB). 2. The system displays a message: "File is too large. The maximum size is 2MB." 3. The user must select a smaller file.
A3: User cancels editing	1. The user clicks the cancel button. 2. The system discards all changes and returns to the profile page.

7. Attraction Explorer Module - Use Case Diagram & Description (Kiang Min Wei)

7.1 Use Case Diagram



7.2 Use Case Description

UC-AE-01: Search Attractions

Item	Description
Use Case ID	UC-AE-01
Use Case Name	Search Attractions
Primary Actor	Guest, Registered User
Description	This use case allows users to search for tourist attractions in Melaka by entering keywords such as attraction name, location, or attraction category.
Precondition	The user has accessed the Chatlas PWA and is on the homepage.
Postcondition	A list of matching attractions is displayed to the user.

Main Flow	
Guest / Registered User	System
1. Opens the Chatlas homepage	2. Displays the homepage with a search bar and a list of attractions.
3. Enters a keyword into the search bar	
4. Clicks the search button or submits the keyword.	5. Receives and Processes the keyword and searches for matching attractions.
	6. Redirects the users to the Search Result Page.
	7. Displays attractions that match the keyword.

Alternative Flow	
Alternative Flow	Description
A1: No matching attraction found	1. The user enters a keyword that does not match any attraction. 2. The system redirects the user to the Search Result Page. 3. The system displays a message such as “No attraction found.” 4. The user may enter another keyword or return to the homepage.
A2: User applies filter	1. The user is viewing the Search Result Page. 2. The user opens the filter option. 3. The system continues to UC-AE-02: Filter Attractions.
A3: User selects attraction details	1. The user selects an attraction from the search results. 2. The system opens the selected attraction detail page. 3. The system continues to UC-AE-03: View Attraction Details.

UC-AE-02: Filter Attractions

Item	Description
Use Case ID	UC-AE-02
Use Case Name	Filter Attractions
Primary Actor	Guest, Registered User
Description	This use case allows users to narrow down attraction search results by applying filters such as attraction category, location area or rating.
Precondition	The user is viewing the Search Result Page.
Postcondition	The attraction results are updated based on the selected filters.

Main Flow	
Guest / Registered User	System
1. Views the Search Result Page.	2. Displays the attraction search results.
3. Open the filter option.	4. Displays available filter criteria such as attraction category, location area or rating.
5. Selects one or more filters.	6. Receives and processes the selected filter criteria.
	7. Displays the updated attraction results based on the selected filters.
8. View the filtered attraction results.	

Alternative Flow	
Alternative Flow	Description
A1: No result after filtering	1. The user selects filters that do not match any attraction. 2. The system displays a message such as “No attraction matches the selected filters.” 3. The user may change the filter selection or reset all filters.
A2: User cancels filtering	1. The user opens the filter option. 2. The user closes the filter option without applying any filter. 3. The system keeps the original search results unchanged.
A3: User selects attraction details	1. The user selects an attraction from the filtered results. 2. The system opens the selected attraction detail page. 3. The system continues to UC-AE-03: View Attraction Details.

UC-AE-03: View Attraction Details

Item	Description
Use Case ID	UC-AE-03
Use Case Name	View Attraction Details
Primary Actor	Guest, Registered User
Description	This use case allows users to view detailed information about a selected attraction, including attraction name, description, category, address, rating, and photos.
Precondition	The user has selected an attraction from the homepage attraction list, Search Result Page, or filtered results.
Postcondition	The selected attraction detail page is displayed to the user.

Main Flow	
Guest / Registered User	System
1. Selects an attraction from the homepage attraction list, Search Result Page, or filtered results.	2. Receives the selected attraction request.
	3. Retrieves the selected attraction information.
	4. Displays the attraction detail page with attraction name, description, category, address, rating and photos (UC-AE-04).
5. Read the attraction information.	6. Keeps the attraction detail page available for viewing.

Alternative Flow	
Alternative Flow	Description
A1: Attraction details cannot be loaded	1. The user selects an attraction. 2. The system fails to retrieve the attraction details due to network or data issues. 3. The system displays an error message. 4. The user may refresh the page or return to the previous page.
A2: User wants to view location	1. The user selects the map or location option on the attraction detail page. 2. The system opens the attraction location map. 3. The system continues to UC-AE-05: View Attraction Location (Map).

UC-AE-04: View Attraction Photos

Item	Description
Use Case ID	UC-AE-04
Use Case Name	View Attraction Photos
Primary Actor	Guest, Registered User
Description	This use case allows users to view available photo previews of a selected attraction on the attraction detail page. The user may optionally select a photo to view it in a larger size.
Precondition	The user is viewing the attraction detail page.
Postcondition	The attraction photos are displayed to the user, or a clear message is shown if no photos are available.

Main Flow	
Guest / Registered User	System
	1. Loads photos related to the selected attraction.
	2. Check whether attraction photos are available.
	3. Displays the available photo previews on the attraction detail page.
4. View the photo previews.	

Alternative Flow	
Alternative Flow	Description
A1: No photo available	1. The user opens the attraction detail page. 2. The system cannot find any photo for the selected attraction. 3. The system displays a default placeholder image or a message such as "No photo available."
A2: User enlarges an attraction photo	1. The user selects an available photo preview. 2. The system opens the selected photo in a larger view. 3. The user views the selected photo. 4. The user closes the larger photo view. 5. The system returns to the attraction detail page.
A3: User closes photo preview	1. The user closes the enlarged photo view. 2. The system returns the user to the attraction detail page.

UC-AE-05: View Attraction Location (Map)

Item	Description
Use Case ID	UC-AE-05
Use Case Name	View Attraction Location (Map)
Primary Actor	Guest, Registered User
Description	This use case allows users to optionally view the selected attraction's location on an interactive map from the attraction detail page. The system displays the attraction address and places a marker on the map. This function does not include route navigation or real-time direction guidance.
Precondition	The user is viewing the attraction detail page.
Postcondition	The attraction location is displayed on the map, or the attraction address is shown if the map cannot be loaded.

Main Flow	
Guest / Registered User	System
1. Selects the map or location option on the attraction detail page.	2. Receives the request to view the attraction location.
	3. Retrieves the attraction's location information.
	4. Displays an interactive map with a marker on the selected attraction.
5. Views the attraction location on the map.	
6. Check the nearby area around the attraction.	7. Allows the user to view the attraction's surrounding area on the map.

Alternative Flow	
Alternative Flow	Description
A1: Map cannot be loaded	1. The user selects the map or location option. 2. The system fails to load the map due to network issues or unavailable map service. 3. The system displays the attraction address in text form. 4. The system shows a message such as "Map is currently unavailable."
A2: Location data unavailable	1. The user attempts to view the attraction location. 2. The system cannot find the location data for the selected attraction. 3. The system displays a message such as "Location information is currently unavailable."
A3: User closes map view	1. The user closes the map view. 2. The system returns the user to the attraction detail page.

UC-AE-06: Browse Attractions

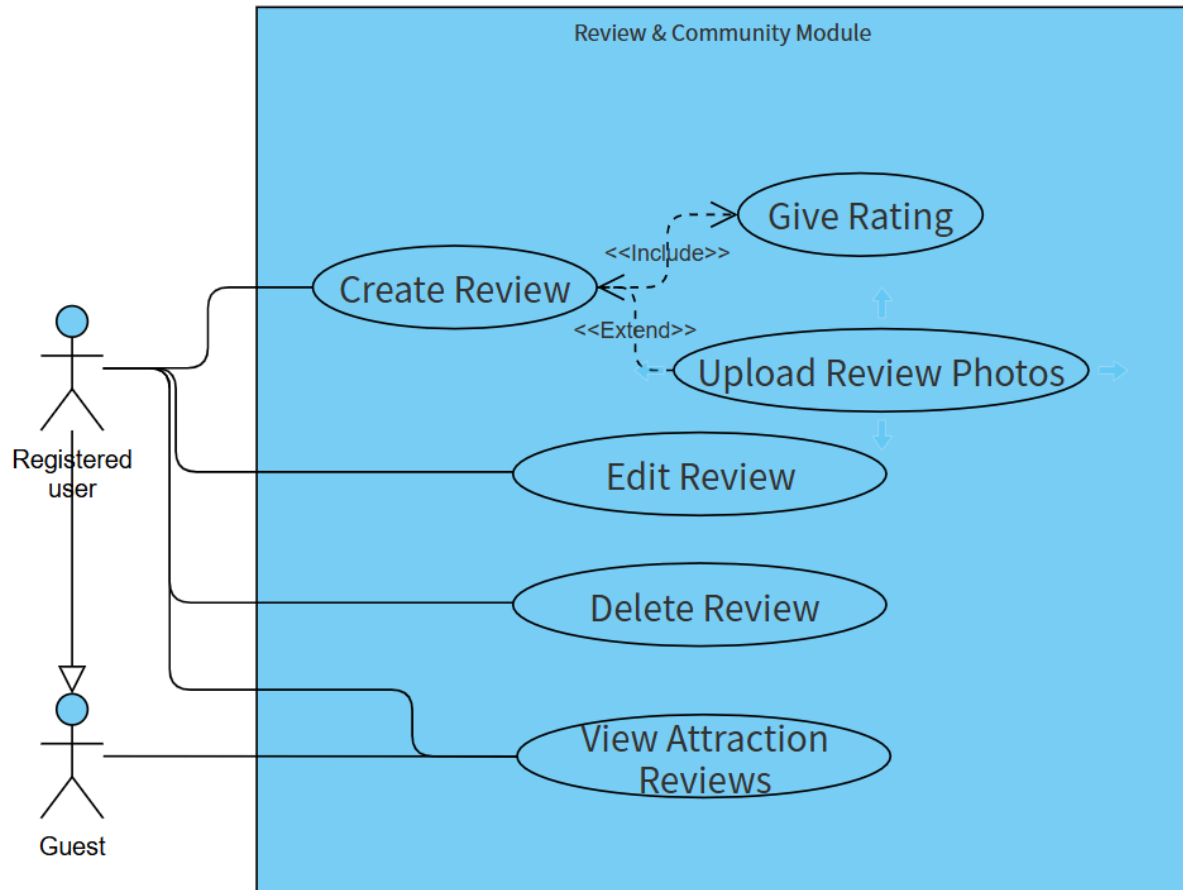
Item	Description
Use Case ID	UC-AE-06
Use Case Name	Browse Attractions
Primary Actor	Guest, Registered User
Description	This use case allows users to browse tourist attractions in Melaka from the homepage without entering a search keyword
Precondition	The user has accessed the Chatlas PWA and is on the homepage
Postcondition	The available attraction list is displayed for browsing

Main Flow	
Guest / Registered User	System
1. Open the Chatlas homepage	2. Retrieves the available attraction information.
	3. Display a list of attractions with basic summary information.
4. Scroll through and browse the attraction list.	5. Keeps the attraction list available for browsing.

Alternative Flow	
Alternative Flow	Description
A1: User selects an attraction	1. The user selects an attraction from the displayed list. 2. The system opens the selected attraction detail page. 3. The system continues to UC-AE-03: View Attraction Details.
A2: No attractions are available	1. The system cannot find any available attraction records. 2. The system displays a message such as “No attractions are currently available.”
A3: Attraction list cannot be loaded	1. The system fails to retrieve the attraction list due to a network or data issue. 2. The system displays an error message. 3. The user may refresh the page and try again.
A4: User decides to search	1. The user selects the search bar and enters a keyword. 2. The system continue to UC-AE-01: Search Attractions.

8. Review Community Module - Use Case Diagram & Description (Tee Jun Jeff)

8.1 Use Case Diagram



8.2 Use Case Description

UC-RC-01: Create Review

Item	Description
Use Case ID	UC-RC-01
Use Case Name	Create Review
Primary Actor	Registered User
Description	This use case allows authenticated users to share their travel experiences by writing text commentary for a specific tourist attraction in Melaka. This use case includes selecting a rating score and optionally extending to upload media photos
Precondition	The user is logged into their Chatlas account and is viewing a specific attraction detail page.
Postcondition	The written review data is saved in the database, published to the community feed, and mapped to the user's travel footprint.

Main Flow	
Registered User	System
1. Click on the "Write a Review" button on the attraction detail page.	2. Displays the review creation form containing a text area, rating stars, and image attachment slot.
3. Selects a star rating for the attraction.	4. Includes UC-RC-02: Give Rating to capture and validate the numerical score.
5. Enter their written feedback in the text field.	
6. Submit the form by clicking the "Submit Review" button.	7. Verifies that the required text field input is valid and not empty.
	8. Saves the review to the MongoDB database and recalculates the attraction's overall average rating
	9. Displays a success confirmation alert and dynamically updates the user's exploration map progress.

Alternative Flow	
Alternative Flow	Description
A1: Empty text submission	1. The user attempts to submit the review form with a blank text area. 2. The system blocks the submission transaction. 3. The system highlights the text box and prompts: "Review text cannot be empty."
A2: User attaches photos	1. The user decides to add visual proof of their visit before step 6. 2. The system branches and continues to UC-RC-03: Upload Review Photos
A3: Network lost during submission	1. The system detects a total drop in connectivity when the user submits. 2. The backend write fails; the system redirects the user to a graceful offline notification screen to preserve input state.

UC-RC-02: Giving Rating

Item	Description
Use Case ID	UC-RC-02
Use Case Name	Give Rating
Primary Actor	Registered User
Description	This use case captures a user's numerical 1-to-5 star evaluation score for an attraction. It is a mandatory included process for creating or editing reviews.
Precondition	The user has initialized the review submission form interface via UC-RC-01 or UC-RC-04
Postcondition	A valid integer score is saved and bound to the pending review form payload.

Main Flow	
Registered User	System
	1. Renders five interactive star icon vectors inside the active review workspace.
2. Hover or taps across the star selection scale to define their rating level.	3. Listens to the user click event and highlights the selected number of star objects sequentially on-screen..
	4. Binds the corresponding integer metric (1 for Poor up to 5 for Excellent) into the pending active review object.
	5. Returns execution control back to the invoking master flow step.

Alternative Flow	
Alternative Flow	Description
A1: Zero rating validation	1. The user bypasses selecting any star level and proceeds to final review submission. 2. The system captures an unselected rating error condition. 3. The system blocks submission, flashes an validation text stating "Rating selection is required", and focuses back onto the star interaction component.

UC-RC-03: Upload Review Photos

Item	Description
Use Case ID	UC-RC-03
Use Case Name	Upload Review Photos
Primary Actor	Registered User
Description	This use case allows users to optionally browse and upload local image files from their personal device storage to support a tourist review entry.
Precondition	The user is actively operating within the review composition form interface.
Postcondition	Selected visual media attachments are verified, staged locally, and ready for database persistence.

Main Flow	
Registered User	System
1.Clicks on the "Add Photos" upload box layout.	2.Triggers the client device's native file selection explorer dialog or image gallery app.
3.Chooses specific media files and confirms selection.	4. Intercepts files and validates matching media parameters (.jpg, .png file types; size limit $\leq$ 5MB per file).
	5. Instantiates local base64 preview thumbnails inside the form layout workspace.
	6. Appends verified image file objects into the review submission payload and returns control to the master form flow.

Alternative Flow	
Alternative Flow	Description
A1: File type / size validation error	1. The user selects a document type (e.g., pdf) or an image exceeding 5MB. 2. The system automatically screens and rejects the invalid attachment file. 3. The system renders an error message box noting file criteria rules, leaving the upload staging container clean.

UC-RC-04: Edit Review

Item	Description
Use Case ID	UC-RC-04
Use Case Name	Edit Review
Primary Actor	Registered User
Description	This use case allows a logged-in user to modify the text content, rating level, or attached photos of a review they previously authored.
Precondition	The user is logged in, online, and viewing a review component where they are verified as the original author.
Postcondition	The modified review data updates the target document entry inside the database collection.

Main Flow	
Registered User	System
1.Taps on the "Edit" action button overlay located on their historical review container card.	2. Pulls target review values from the database and maps them back into an editable form layout.
3.Modifies text blocks, changes rating selections, or updates uploaded photos.	4. Includes UC-RC-02: Give Rating if star adjustments occur.
5. Submits modifications by clicking the "Update Review" button.	6. Validates newly revised input values for structural completeness.
	7. Submits a PUT update query to overwrite historical field coordinates in MongoDB.
	8. Triggers an automated update to the attraction's global rating parameters and displays an "Update Successful" notice message.

Alternative Flow	
Alternative Flow	Description
A1: User clears text completely	1. The user completely erases the text body and clicks update. 2. The system triggers empty validation safeguards. 3. The system halts the operation, displaying a validation note stating review text is required.

UC-RC-04: Edit Review

Item	Description
Use Case ID	UC-RC-05
Use Case Name	Delete Review
Primary Actor	Registered User
Description	This use case allows an authenticated user to permanently erase their published review, dropping it entirely from community feeds and attraction score matrices.
Precondition	The user is logged in, online, and viewing a review card they originally authored.
Postcondition	The review document is permanently removed from MongoDB, altering community totals and map footprint syncs.

Main Flow	
Registered User	System
1. Click the "Delete Review" button options dropdown on their specific review card.	2. Pauses view focus to construct a confirmation modal prompt asking: "Are you sure you want to permanently delete this review?".
3. Click on the "Confirm Delete" action link inside the modal.	4. Dispatches an authorized DELETE request action pipeline to the Node.js API instance.
	5. Hard-deletes the specific review record document directly from MongoDB
	6. Adjusts attraction average values and updates Module 4 to evaluate if the site should be unmarked from the user's Exploration Map.
	7. Closes the overlay modal and dynamically removes the review card container layout from the viewport feed.

Alternative Flow	
Alternative Flow	Description
A1: User cancels deletion request	1. The user hits the "Cancel" option link inside the verification pop-up framework. 2. The system tears down the confirmation modal instantly. 3. The system makes zero data mutation calls, leaving review states un-altered.

UC-RC-06: Edit Review

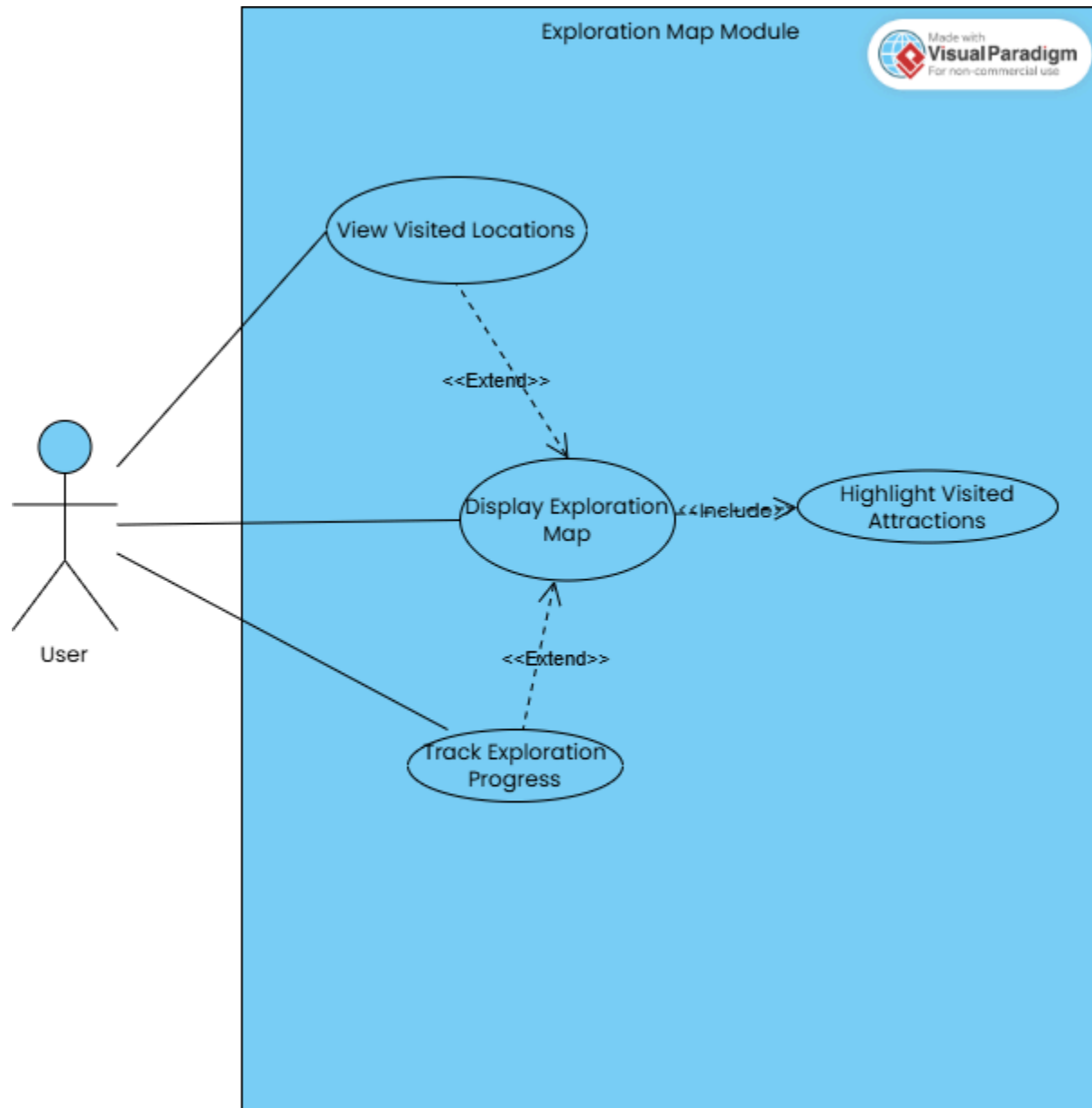
Item	Description
Use Case ID	UC-RC-05
Use Case Name	View Attraction Reviews
Primary Actor	Guest, Registered User
Description	This use case allows any visitor to read community reviews, review ratings, and browse uploaded photos linked to a selected attraction in Melaka.
Precondition	The user has navigated onto an attraction details page layout inside Chatlas PWA.
Postcondition	A sorted timeline array of crowd-sourced review cards is successfully displayed to the user.

Main Flow	
Registered User	System
1. Navigates down to the review section area of the active attraction profile page.	2. Parses the active attraction tracking ID path variable parameter.
	3. Executes a query request to locate matching review collection entries.
	4. Formats and organizes reviews chronologically.
	5. Renders average score totals, author profiles, review texts, and uploaded photo grids onto the display viewport.

Alternative Flow	
Alternative Flow	Description
A1: Zero community reviews found	1. The database checks and confirms zero review entries are registered for the target site. 2. The system suppresses timeline component layouts. 3. The system presents a clean placeholder notice saying: "No reviews yet. Be the first to share your experience!".
A2: Device is offline (PWA Caching Mode)	1. The system captures that the app is running with zero network connection. 2. The PWA Service Worker intercepts backend requests and pulls previously viewed review structures out from Cache Storage. 3. The system displays cached data on the screen with an "Offline Mode Cached View" badge indicator.

9. Exploration Map Module - Use Case Diagram & Description (Soh Lee Xuan)

9.1 Use Case Diagram



9.2 Use Case Description

UC-EM-01: Display Exploration Map

Item	Description
Use Case ID	UC-EM-01
Use Case Name	Display Exploration Map
Primary Actor	Registered User
Description	This use case allows a registered user to view their personal exploration map for Melaka. The map displays supported attractions and provides a visual overview of the user's exploration activity.
Precondition	The user has logged in to the Chatlas PWA and selects the Exploration Map module.
Postcondition	The personal exploration map is displayed to the user.

Main Flow

Registered User	System
1. Opens the Exploration Map module.	2. Verifies that the user is logged in.
3. Requests to view the personal exploration map.	4. Retrieves the Melaka map data and attraction location data.
	5. Retrieves the user's visited attraction records.
	6. Displays the personal exploration map.
	7. Continues to UC-EM-02: Highlight Visited Attractions.
8. Views the exploration map.	9. Keeps the exploration map available for viewing.

Alternative Flow

Alternative Flow	Description
A1: User has no visited attractions	1. The system finds no visited attraction records for the user. 2. The system displays the exploration map without highlighted attractions. 3. The system shows a message such as “No visited attractions yet.”
A2: Map cannot be loaded	1. The system fails to load the map due to network or map service issues. 2. If cached map data is available, the system displays the cached map region. 3. If cached data is unavailable, the system displays a message such as “Map is currently unavailable.”
A3: User session expired	1. The system detects that the user session has expired. 2. The system redirects the user to the login page. 3. The user must log in again before accessing the exploration map.

UC-EM-02: Highlight Visited Attractions

Item	Description
Use Case ID	UC-EM-02
Use Case Name	Highlight Visited Attractions
Primary Actor	Registered User
Description	This use case allows the system to automatically highlight attractions that the user has visited on the exploration map.
Precondition	The user is viewing the exploration map, and the system has access to the user's visited attraction records.
Postcondition	Visited attractions are visually highlighted on the exploration map.

Main Flow

Registered User	System
1. Views the exploration map.	2. Retrieves the user's visited attraction records.
	3. Matches the visited attractions with attraction location data.
	4. Applies a highlighted marker or visual indicator to visited attractions.
	5. Displays highlighted visited attractions on the map.
6. Views the highlighted attractions.	7. Keeps the highlighted map available for viewing.

Alternative Flow

Alternative Flow	Description
A1: No visited attraction found	1. The system cannot find any visited attraction records for the user. 2. The system displays the map without highlighted markers. 3. The system shows a message such as “Start exploring to highlight your visited attractions.”
A2: Location data unavailable	1. The system finds a visited attraction record but cannot locate its map coordinates. 2. The system skips the map highlight for that attraction. 3. The attraction may still appear in the visited locations list.
A3: Visited records cannot be retrieved	1. The system fails to retrieve the user’s visited records due to network or data issues. 2. The system displays the map without highlighted attractions. 3. The system shows an error message such as “Visited attractions cannot be loaded.”

UC-EM-03: View Visited Locations

Item	Description
Use Case ID	UC-EM-03
Use Case Name	View Visited Locations
Primary Actor	Registered User
Description	This use case allows a registered user to view the list of attractions they have visited in Melaka.
Precondition	The user has logged in and is viewing the Exploration Map module.
Postcondition	The user's visited locations are displayed.

Main Flow

Registered User	System
1. Opens the Exploration Map module.	2. Displays the exploration map.
3. Selects the "Visited Locations" option or clicks a highlighted attraction marker.	4. Retrieves the user's visited attraction records.
	5. Displays the list of visited locations.
6. Views the visited locations list.	7. Displays basic information such as attraction name, category, and location.
8. Selects a visited location.	9. Displays the selected visited location summary.

Alternative Flow

Alternative Flow	Description
A1: No visited locations available	1. The user selects the visited locations option. 2. The system finds no visited location records. 3. The system displays a message such as “No visited locations yet.”
A2: Visited locations cannot be loaded	1. The system fails to retrieve the visited locations due to network or data issues. 2. The system displays an error message. 3. The user may refresh the page or return to the exploration map.
A3: User closes visited locations list	1. The user closes the visited locations list. 2. The system returns the user to the exploration map.

UC-EM-04: Track Exploration Progress

Item	Description
Use Case ID	UC-EM-04
Use Case Name	Track Exploration Progress
Primary Actor	Registered User
Description	This use case allows a registered user to view their exploration progress based on the number of attractions they have visited in Melaka.
Precondition	The user has logged in and has accessed the Exploration Map module.
Postcondition	The user's exploration progress is calculated and displayed.

Main Flow

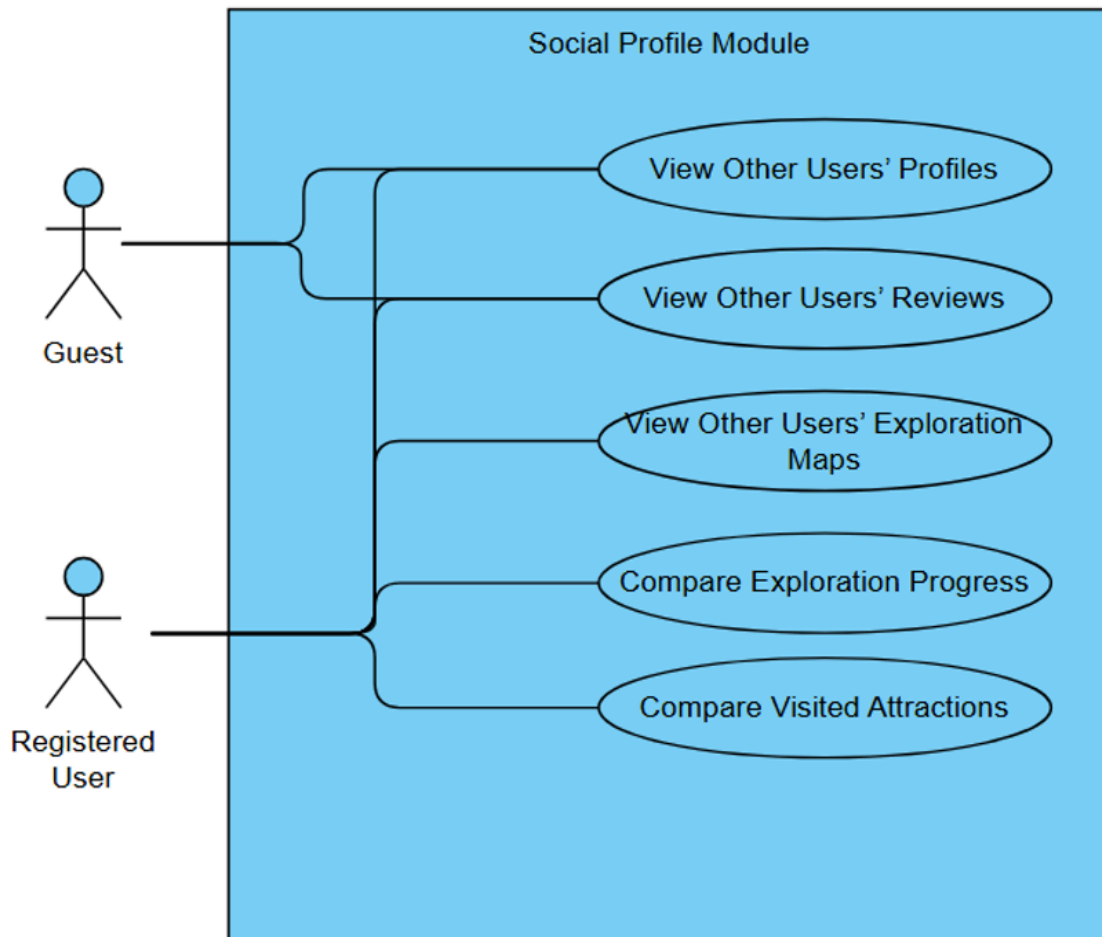
Registered User	System
1. Opens the Exploration Map module.	2. Displays the exploration map.
3. Selects the "Exploration Progress" option.	4. Retrieves the total number of supported attractions in Melaka.
	5. Retrieves the number of attractions visited by the user.
	6. Calculates the user's exploration progress.
	7. Displays the progress summary, such as visited attraction count and completion percentage.
8. Views the exploration progress.	9. Keeps the progress summary available for viewing.

Alternative Flow

Alternative Flow	Description
A1: User has not visited any attraction	1. The system finds no visited attraction records. 2. The system displays the progress as 0%. 3. The system shows a message such as “No exploration progress yet.”
A2: Total attraction data unavailable	1. The system cannot retrieve the total number of supported attractions. 2. The system displays only the number of visited attractions. 3. The system hides or disables the completion percentage.
A3: Progress cannot be calculated	1. The system fails to calculate the exploration progress due to data issues. 2. The system displays an error message. 3. The user may refresh the page or return to the exploration map.

10. Social Profile Module - Use Case Diagram & Description (Leong Yong Quan)

10.1 Use Case Diagram



10.2 Use Case Description

UC-SP-01: View Other Users' Profiles

Item	Description
Use Case ID	UC-SP-01
Use Case Name	View Other Users' Profiles
Primary Actor	Guest, Registered User
Description	This use case allows users to view another user's public social profile, including basic profile information and public travel activity summary.
Precondition	The user has accessed the Chatlas PWA and the target user profile exists in the system.
Postcondition	The selected user's public profile information is displayed to the user.

Main Flow

Guest / Registered User	System
1. Opens the Chatlas PWA.	2. Displays the homepage or main navigation page.
3. Opens the Social Profile section or selects a user profile from the platform.	4. Receives the request and searches for the selected user profile.
	5. Retrieves the selected user's public profile information.
	6. Displays the selected user's public profile page.
7. Views the other user's public profile information.	

Alternative Flow

Alternative Flow	Description
A1: Target user profile not found	1. The user selects or searches for a user profile that does not exist. 2. The system displays a message such as "User profile not found." 3. The user may search for another user or return to the previous page.
A2: User selects reviews section	1. The user is viewing another user's profile page. 2. The user selects the reviews section. 3. The system continues to UC-SP-02: View Other Users' Reviews.
A3: User selects exploration map section	1. The user is viewing another user's profile page. 2. The user selects the exploration map section. 3. The system continues to UC-SP-03: View Other Users' Exploration Maps.
A4: Registered user selects comparison option	1. The registered user is viewing another user's profile page. 2. The registered user selects the comparison option. 3. The system continues to UC-SP-04: Compare Exploration Progress or UC-SP-05: Compare Visited Attractions.

UC-SP-02: View Other Users' Reviews

Item	Description
Use Case ID	UC-SP-02
Use Case Name	View Other Users' Reviews
Primary Actor	Guest, Registered User
Description	This use case allows users to view reviews written by another user, including review content, ratings, reviewed attractions, and uploaded review photos.
Precondition	The user has accessed the Chatlas PWA and the target user profile exists in the system.
Postcondition	The selected user's public reviews are displayed to the users.

Main Flow

Guest / Registered User	System
1. Opens the Chatlas PWA.	2. Displays the homepage or main navigation page.
3. Opens another user's social profile page.	4. Retrieves the selected user's public profile information.
5. Selects the reviews section on the profile page.	6. Receives the request and searches for reviews created by the selected user.
	7. Retrieves the selected user's public review records, including ratings, review text, attraction names, and review photos.
	8. Displays the selected user's reviews on the profile page.
9. Views the other user's reviews.	

Alternative Flow

Alternative Flow	Description
A1: Target user has no public reviews	1. The users selects the reviews section of another user's profile. 2. The system searches for public reviews created by the selected user. 3. No review records are found. 4. The system displays a message such as "No reviews available."
A2: Review photos are unavailable	1. The system retrieves the selected user's public reviews. 2. Some review photos are missing, removed, or unavailable. 3. The system displays the review text and rating without the unavailable photos.
A3: Users select an attraction from a review	1. The user is viewing another user's review. 2. The users selects the attraction name shown in the review. 3. The system continues to the Attraction Explorer module to display the selected attraction details.
A4: Users return to profile page	1. The user is viewing another user's reviews. 2. The user selects the back option. 3. The system returns to UC-SP-01: View Other Users' Profiles.

UC-SP-03: View Other Users' Exploration Maps

Item	Description
Use Case ID	UC-SP-03
Use Case Name	View Other Users' Exploration Maps
Primary Actor	Registered User
Description	This use case allows a registered user to view another user's public exploration map, including visited attractions and exploration progress based on the user's reviewed or visited attractions.
Precondition	The registered user has logged in using a Google account and the target user profile exists in the system.
Postcondition	The selected user's public exploration map is displayed to the registered user.

Main Flow

Registered User	System
1. Opens the Chatlas PWA.	2. Displays the homepage or main navigation page.
3. Opens another user's social profile page.	4. Retrieves the selected user's public profile information.
5. Selects the exploration map section on the profile page.	6. Receives the request and searches for the selected user's exploration map data.
	7. Retrieves the selected user's visited attraction records and exploration progress data.
	8. Displays the selected user's exploration map with highlighted visited attractions.
9. Views the other user's exploration map and visited locations.	

Alternative Flow

Alternative Flow	Description
A1: Target user has no visited attraction records	1. The registered user selects the exploration map section of another user's profile. 2. The system searches for the selected user's visited attraction records. 3. No visited attraction records are found. 4. The system displays an empty exploration map with a message such as "No visited attractions available."
A2: Exploration map data is unavailable	1. The system attempts to retrieve the selected user's exploration map data. 2. The map data cannot be loaded due to a system or connection issue. 3. The system displays a message such as "Exploration map is currently unavailable."
A3: Registered user selects a visited attraction on the map	1. The registered user is viewing another user's exploration map. 2. The registered user selects a visited attraction marker on the map. 3. The system continues to the Attraction Explorer module to display the selected attraction details.
A4: Registered user returns to profile page	1. The registered user is viewing another user's exploration map. 2. The registered user selects the back option. 3. The system returns to UC-SP-01: View Other Users' Profiles.

UC-SP-04: Compare Exploration Progress

Item	Description
Use Case ID	UC-SP-04
Use Case Name	Compare Exploration Progress
Primary Actor	Registered User
Description	This use case allows a registered user to compare their exploration progress with another user, including the number or percentage of visited attractions.
Precondition	The registered user has logged in using a Google account, and the target user profile exists in the system.
Postcondition	The comparison result of exploration progress between the registered user and the selected user is displayed.

Main Flow

Registered User	System
1. Opens the Chatlas PWA.	2. Displays the homepage or main navigation page.
3. Opens another user's social profile page.	4. Retrieves the selected user's public profile information.
5. Select the compare exploration progress option.	6. Receives the request and retrieves the registered user's exploration progress data.
	7. Retrieves the selected user's exploration progress data.
	8. Calculates and compares both users' exploration progress.
	9. Displays the comparison result, such as visited attraction count or exploration percentage.
10. Views the exploration progress comparison result.	

Alternative Flow

Alternative Flow	Description
A1: Guest selects the comparison option	1. The guest selects the compare exploration progress option. 2. The system displays a message such as "Please log in with Google to compare exploration progress." 3. The guest may proceed to the Google login process.
A2: Target user has no exploration progress	1. The registered user selects the compare exploration progress option. 2. The system retrieves the selected user's exploration progress data. 3. No exploration progress data is found for the selected user. 4. The system displays a message such as "This user has no exploration progress yet."
A3: Registered user has no exploration progress	1. The registered user selects the compare exploration progress option. 2. The system retrieves the registered user's exploration progress data. 3. No exploration progress data is found for the registered user. 4. The system displays a message such as "You have no exploration progress yet."
A4: Registered user selects visited attraction comparison	1. The registered user is viewing the exploration progress comparison result. 2. The registered user selects the visited attraction comparison option. 3. The system continues to UC-SP-05: Compare Visited Attractions.
A5: Registered user returns to profile page	1. The registered user is viewing the exploration progress comparison result. 2. The registered user selects the back option. 3. The system returns to UC-SP-01: View Other Users' Profiles.

UC-SP-05: Compare Visited Attractions

Item	Description
Use Case ID	UC-SP-05
Use Case Name	Compare Visited Attractions
Primary Actor	Registered User
Description	This use case allows a registered user to compare their visited attractions with another user to identify common visited attractions and different visited attractions.
Precondition	The registered user has logged in using a Google account, and the target user profile exists in the system.
Postcondition	The comparison result of visited attractions between the registered user and the selected user is displayed.

Main Flow

Registered User	System
1. Opens the Chatlas PWA.	2. Displays the homepage or main navigation page.
3. Opens another user's social profile page.	4. Retrieves the selected user's public profile information.
5. Selects the compare visited attractions option.	6. Receives the request and retrieves the registered user's visited attraction records.
	7. Retrieves the selected user's visited attraction records.
	8. Compare both user's visited attraction records.
	9. Identifies common visited attractions and attractions visited by only one user.
	10. Displays the visited attraction comparison result.
11. Views the comparison result of visited attractions.	

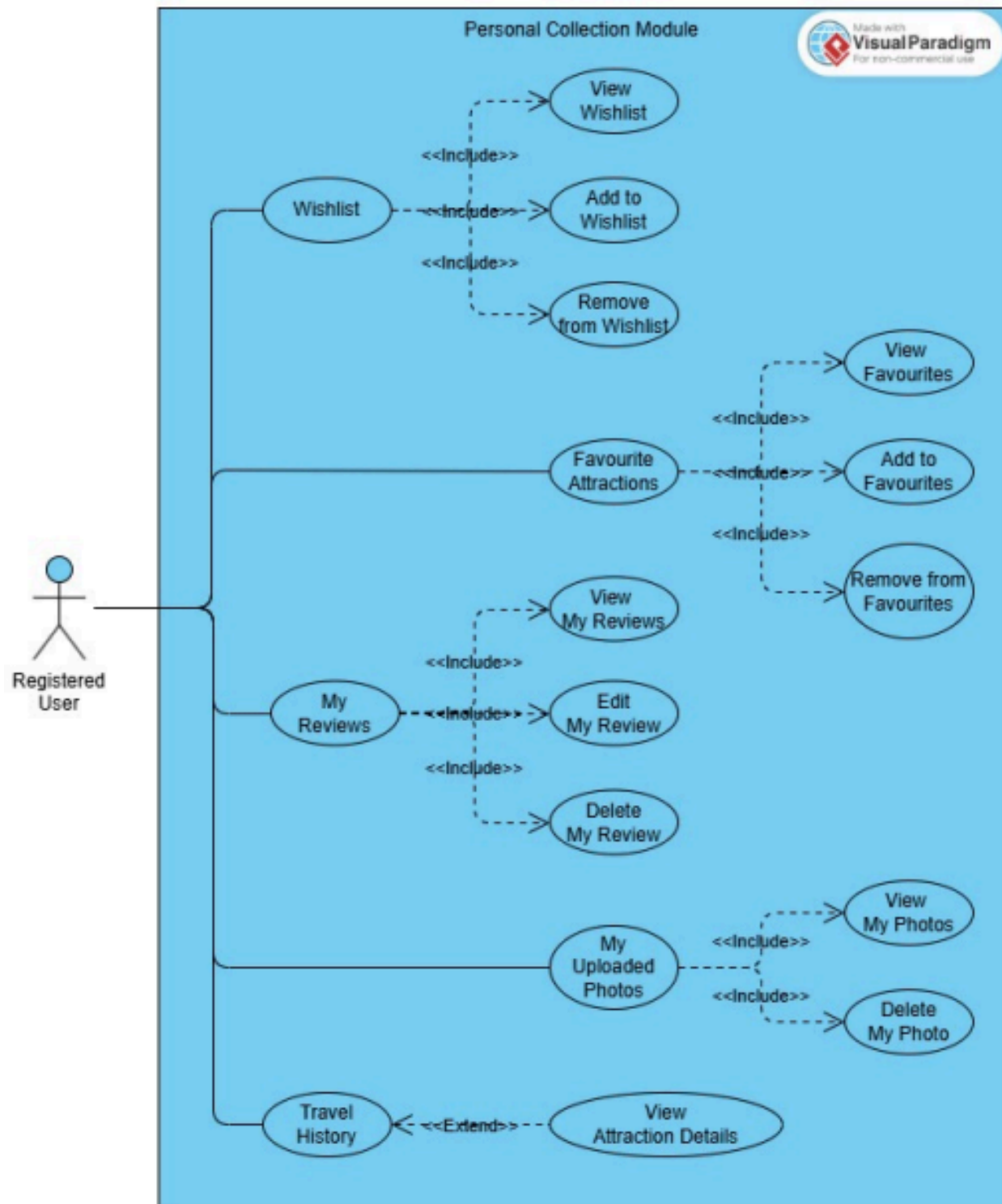
Alternative Flow

Alternative Flow	Description
A1: Guest selects the comparison option	1. The guest selects the compare visited attractions option. 2. The system displays a message such as "Please log in with Google to compare visited attractions." 3. The guest may proceed to the Google login process.
A2: Target user has no visited attraction records	1. The registered user selects the compare visited attractions option. 2. The system retrieves the selected user's visited attraction records. 3. No visited attraction records are found for the selected user. 4. The system displays a message such as "This user has not visited attractions yet."
A3: Registered user has no visited attraction records	1. The registered user selects the compare visited attractions option. 2. The system retrieves the registered user's visited attraction records. 3. No visited attraction records are found for the registered user. 4. The system displays a message such as "You have not visited attractions yet."
A4: No common visited attractions found	1. The system compares both user's visited attraction records. 2. No common visited attractions are found. 3. The system displays a message such as "No common visited attractions found." 4. The system still displays attractions visited by each user separately.
A5: Registered user selects an attraction from the comparison result	1. The registered user is viewing the visited attraction comparison result. 2. The registered user selects an attraction from the list. 3. The system continues to the Attraction Explorer module to display the selected attraction details.

A6: Registered user returns to profile page	1. The registered user is viewing the visited attraction comparison result. 2. The registered user selects the back option. 3. The system returns to UC-SP-01: View Other Users' Profiles.
--	--

11. Personal Collection Module - Use Case Diagram & Description (Ruvy Chong Rui Hui)

11.1 Use Case Diagram



11.2 Use Case Description

UC-PC-01: Wishlist

Item	Description
Use Case ID	UC-PC-01
Use Case Name	Wishlist
Primary Actor	Registered User
Description	This use case allows users to manage their wishlist by viewing all saved attractions, adding new attractions to the wishlist, and removing attractions from the wishlist.
Precondition	The user is logged in.
Postcondition	The user's wishlist is updated based on their actions.

Main Flow	
Registered User	System
1. The user navigates to the "My Wishlist" page.	
	2. Retrieves and displays all attractions in the user's wishlist.
3. The user views the wishlist.	
4. The user selects an action: add an attraction or remove an attraction.	
	5. Processes the user's selected action and updates the wishlist accordingly.
6. The user sees the updated wishlist.	

Alternative Flow	
Alternative Flow	Description
A1: Empty wishlist	1. The user navigates to "My Wishlist" but has not saved any attractions. 2. The system displays a message: "Your wishlist is empty. 3. Start exploring attractions to add to your wishlist!"
A2: Add attraction	1. The user selects the "Save to Wishlist" button on an attraction detail page. 2. The system adds the attraction and displays a confirmation message: "Attraction added to wishlist."
A3: Remove attraction	1. The user selects the "Remove" button next to an attraction in the wishlist. 2. The system removes the attraction and displays a confirmation message: "Attraction removed from wishlist."
A4: Attraction already in wishlist	1. The user attempts to add an attraction already in their wishlist. 2. The system displays a message: "Attraction is already in your wishlist."
A5: Data loading error	1. The system fails to retrieve the wishlist data. 2. The system displays an error message: "Unable to load wishlist. Please refresh the page."

UC-PC-02: Favourite Attractions

Item	Description
Use Case ID	UC-PC-02
Use Case Name	Favourite Attractions
Primary Actor	Registered User
Description	This use case allows users to manage their favourite attractions by viewing all favourite attractions, adding new attractions to favourites, and removing attractions from favourites.
Precondition	The user is logged in.
Postcondition	The user's favourite attractions list is updated based on their actions.

Main Flow	
Registered User	System
1. The user navigates to the "My Favourites" page.	
	2. Retrieves and displays all attractions in the user's favourites list.
3. The user views the favourite attractions.	
4. The user selects an action: add an attraction or remove an attraction.	
	5. Processes the user's selected action and updates the favourites list accordingly.
6. The user sees the updated favourites list.	

Alternative Flow	
Alternative Flow	Description
A1: Empty favourites list	1. The user navigates to "My Favourites" but has not marked any attractions as favourites. 2. The system displays a message: "You haven't added any attractions to your favourites yet."
A2: Add attraction	1. The user selects the "Add to Favourites" button on an attraction detail page. 2. The system adds the attraction and displays a confirmation message: "Attraction added to favourites."
A3: Remove attraction	1. The user selects the "Remove" button next to an attraction in the favourites list. 2. The system removes the attraction and displays a confirmation message: "Attraction removed from favourites."
A4: Attraction already in favourites	1. The user attempts to add an attraction already in their favourites. 2. The system displays a message: "Attraction is already in your favourites."
A5: Data loading error	1. The system fails to retrieve the favorite data. 2. The system displays an error message: "Unable to load favourites. Please refresh the page."

UC-PC-03: My Reviews

Item	Description
Use Case ID	UC-PC-03
Use Case Name	My Reviews
Primary Actor	Registered User
Description	This use case allows users to manage their reviews by viewing all reviews they have written, editing review content or ratings, and deleting reviews.
Precondition	The user is logged in.
Postcondition	The user's reviews are updated based on their actions.

Main Flow	
Registered User	System
1. The user navigates to the "My Reviews" page.	
	2. Retrieves and displays all reviews written by the user.
3. Users view their reviews.	
4. The user selects an action: edit a review or delete a review.	
	5. Processes the user's selected action and updates the review accordingly.
6. The user sees the updated reviews.	

Alternative Flow	
Alternative Flow	Description
A1: No reviews written	1. The user navigates to "My Reviews" but has not written any reviews. 2. The system displays a message: "You haven't written any reviews yet. Start exploring and share your experiences!"
A2: Edit review	1. The user selects the "Edit" button next to a review, modifies the content or rating, and submits the updated review. 2. The system validates and saves the changes, displaying: "Review updated successfully."
A3: Delete review	1. The user selects the "Delete" button next to a review and confirms deletion. 2. The system permanently deletes the review and displays: "Review deleted successfully."
A4: Validation error	1. The user submits invalid content (e.g., empty content). 2. The system displays an error message: "Please fill in all required fields."
A5: Data loading error	1. The system fails to retrieve the reviews data. 2. The system displays an error message: "Unable to load your reviews. Please refresh the page."

UC-PC-04: My Uploaded Photos

Item	Description
Use Case ID	UC-PC-04
Use Case Name	My Uploaded Photos
Primary Actor	Registered User
Description	This use case allows users to manage their uploaded photos by viewing all photos they have uploaded and deleting unwanted photos.
Precondition	The user is logged in.
Postcondition	The user's uploaded photos are updated based on their actions.

Main Flow	
Registered User	System
1. The user navigates to the "My Photos" page.	
	2. Retrieves and displays all photos uploaded by the user in a gallery view.
3. The user views the photos.	
4. The user selects a photo to view in full size or selects the "Delete" button to remove a photo.	
	5. Processes the user's selected action.
6. The user sees the updated photos.	

Alternative Flow	
Alternative Flow	Description
A1: No photos uploaded	1. The user navigates to "My Photos" but has not uploaded any photos. 2. The system displays a message: "You haven't uploaded any photos yet."
A2: Delete photo	1. The user selects the "Delete" button on a photo and confirms deletion. 2. The system permanently deletes the photo and displays: "Photo deleted successfully."
A3: View full-size photo	1. The user selects a photo thumbnail. 2. The system opens the photo in full size.
A4: User cancels deletion	1. The user is prompted to confirm deletion but cancels. 2. The system returns to "My Photos" without deleting the photo.
A5: Data loading error	1. The system fails to retrieve the photos. 2. The system displays an error message: "Unable to load photos. Please refresh the page."

UC-PC-05: Travel History

Item	Description
Use Case ID	UC-PC-05
Use Case Name	Travel History
Primary Actor	Registered User
Description	This use case allows users to view their complete travel history, including all attractions visited, reviews written, and photos uploaded, organized in chronological order.
Precondition	The user is logged in.
Postcondition	The user's travel history is displayed with a chronological log of all activities.

Main Flow	
Registered User	System
1. The user navigates to the "My Travel History" page.	
	2. Retrieves all activities from the user's profile: visited attractions [UC-EM-02], reviews [UC-RC-01], and photos [UC-RC-03].
	3. Displays the travel history in chronological order with dates, attraction names, and activity types.
4. Users view their travel history.	
5. The user selects an attraction from the history to view its details.	
	6. Opens the selected attraction detail page [UC-AE-03: View Attraction Details].

Alternative Flow	
Alternative Flow	Description
A1: No travel history	1. The user navigates to "My Travel History" but has not explored any attractions. 2. The system displays a message: "You haven't explored any attractions yet. Start your journey today!"
A2: Data loading error	1. The system fails to retrieve the travel history data. 2. The system displays an error message: "Unable to load travel history. Please refresh the page."

12. Project Backlog

ID	User Story	Priority	Story Points
PB01	As a guest, I want to sign in using my Google account so that I can access the registered-user features of Chatlas.	High	5
PB02	As a registered user, I want to log out securely so that other people using the same device cannot access my account.	High	2
PB03	As a registered user, I want to edit my personal profile information so that my profile remains accurate and updated.	Medium	3
PB04	As a registered user, I want to upload or update my profile picture so that I can personalise my profile.	Medium	3
PB05	As a guest, I want to receive a clear message when Google login fails so that I know that I should try again.	Medium	2
PB06	As a registered user, I want the system to handle an expired session safely so that protected pages cannot be accessed without signing in again.	High	3
PB07	As a guest or registered user, I want to browse a default list of Melaka attractions before searching so that I can begin exploring immediately.	High	3
PB08	As a guest or registered user, I want to search attractions by name, location or category so that I can quickly find places that interest me.	High	5
PB09	As a guest or registered user, I want to filter attractions by category, location area and minimum rating so that I can refine the attraction list.	High	5
PB10	As a guest or registered user, I want to combine a search keyword with selected filters so that I can narrow the results more precisely.	High	3
PB11	As a guest or registered user, I want to see the current result count and applied filters so that I understand why particular attractions are displayed.	Medium	2

PB12	As a guest or registered user, I want to clear my keyword or selected filters so that I can restore the default Melaka attraction list.	Medium	2
PB13	As a guest or registered user, I want to receive a helpful no-result state so that I can change my keyword, clear filters or return to the default list.	Medium	2
PB14	As a guest or registered user, I want to view an attraction's name, description, category, address and rating so that I can learn about it before visiting.	High	5
PB15	As a guest or registered user, I want to view photo previews for an attraction so that I can see what the location looks like.	Medium	3
PB16	As a guest or registered user, I want to enlarge a selected attraction photo and use previous, next and close controls so that I can inspect the photos comfortably.	Medium	3
PB17	As a guest or registered user, I want the attraction details to remain usable when no photo is available so that missing images do not block the page.	Medium	2
PB18	As a guest or registered user, I want to view one selected attraction on a map with its name, address, marker and zoom or pan controls so that I can inspect its location.	Medium	3
PB19	As a guest or registered user, I want to see the attraction address in text form when the map is unavailable so that I can still obtain its location information.	Medium	2
PB20	As a guest or registered user, I want previously loaded attraction lists, details, images, searches and map regions to remain viewable during a weak or unavailable connection.	Medium	5
PB21	As a registered user, I want to create a review for an attraction so that I can share my experience with other travellers.	High	5
PB22	As a registered user, I want to give a rating from one to five stars when submitting a review so that I can evaluate the attraction.	High	2

PB23	As a registered user, I want to upload photos together with my review so that other travellers can see my actual experience.	High	3
PB24	As a registered user, I want to edit my own review so that I can correct or update it later.	Medium	3
PB25	As a registered user, I want to delete my own review so that I can remove content that I no longer want to share.	Medium	2
PB26	As a guest or registered user, I want to view reviews posted under an attraction so that I can learn from other travellers.	High	3
PB27	As a registered user, I want clear validation messages for incomplete review text, missing ratings or invalid photos so that I can correct my submission.	Medium	2
PB28	As a user, I want the attraction's average rating to update when valid ratings are submitted so that the displayed rating reflects the available reviews.	High	3
PB29	As a registered user, I want to view my personal exploration map so that I can visualise my travel activity in Melaka.	High	5
PB30	As a registered user, I want attractions that I have reviewed to be highlighted as visited so that my map represents the places I have explored.	High	5
PB31	As a registered user, I want to view a list of my visited attractions so that I can review the places I have explored.	Medium	3
PB32	As a registered user, I want to view my exploration progress so that I know how many supported attractions I have completed.	Medium	3
PB33	As a registered user, I want my exploration map, visited list and progress to update after I submit a review so that all travel records remain consistent.	High	3

PB34	As a registered user, I want clear empty and map-unavailable states so that the Exploration Map remains understandable when I have no visits or the map cannot load.	Medium	2
PB35	As a registered user, I want to view another user's public profile so that I can learn about their travel activities.	Medium	3
PB36	As a registered user, I want to view reviews written by another user from their profile so that I can learn from their experiences.	Medium	3
PB37	As a registered user, I want to view another user's exploration map so that I can see the attractions they have visited.	Medium	5
PB38	As a registered user, I want to compare my exploration progress with another user so that I can understand the difference between our progress.	Medium	5
PB39	As a registered user, I want to compare my visited attractions with another user so that I can identify common and different visited places.	Medium	5
PB40	As a registered user, I want clear user-not-found and no-data states so that I understand when another user has no public reviews or exploration records.	Medium	2
PB41	As a registered user, I want to add and remove attractions from my wishlist so that I can remember places that I may visit in the future.	Medium	3
PB42	As a registered user, I want to add and remove favourite attractions so that I can easily access the places that I like most.	Medium	3
PB43	As a registered user, I want to view all reviews that I have created so that I can manage and revisit my previous reviews.	Medium	3
PB44	As a registered user, I want to view all photos that I have uploaded through my reviews so that I can revisit my shared travel photos.	Medium	3
PB45	As a registered user, I want to view my personal travel history in chronological order so that I can revisit my previous travel activities.	Medium	5

13. Backlog Grooming

ID	User Story	Tasks	Priority	Story Points
PB01	As a guest, I want to sign in using my Google account so that I can access the registered-user features of Chatlas.	Implement the approved Google login page. Configure Google Identity authentication. Create the authentication controller/service and user record handling. Create the authenticated session after successful login. Redirect the user to the correct page after login. Test successful and failed login flows.	High	5
PB02	As a registered user, I want to log out securely so that other people using the same device cannot access my account.	Add the logout action to the approved navigation/profile UI. Clear the active application session. Remove or invalidate authentication tokens. Redirect the user after logout. Block access to protected pages after logout. Test logout and back-navigation behaviour.	High	2
PB03	As a registered user, I want to edit my personal profile information so that my profile remains accurate and updated.	Implement the approved profile and edit-profile pages. Load the current profile information. Validate editable profile fields. Update the user record in the database. Display success and validation messages. Test valid and invalid updates.	Medium	3
PB04	As a registered user, I want to upload or update my profile picture so that I can personalise my profile.	Implement the approved profile-picture upload UI. Validate image type and size. Upload the image to the selected image-storage service. Save the image URL to the user record. Replace the previous profile picture when updated. Handle upload failure and test the flow.	Medium	3
PB05	As a guest, I want to receive a clear message when Google login fails so that I know that I should try again.	Identify expected authentication failure cases. Add the approved login-error UI state. Map provider or network errors to user-friendly messages. Allow the user to retry login. Prevent partial session creation. Test cancelled, failed and unavailable login cases.	Medium	2

PB06	As a registered user, I want the system to handle an expired session safely so that protected pages cannot be accessed without signing in again.	Detects missing or expired sessions on protected routes. Clear invalid local authentication state. Redirect the user to sign in again. Display the approved session-expired message. Prevent protected API operations after expiry. Test expiry during navigation and form submission.	High	3
PB07	As a guest or registered user, I want to browse a default list of Melaka attractions before searching so that I can begin exploring immediately.	Implement the approved integrated Attractions page. Retrieve the default Melaka attraction list. Display attraction cards using the shared design. Add loading and data-unavailable states. Keep the list visible before any search or filter is applied. Test initial page loading.	High	3
PB08	As a guest or registered user, I want to search attractions by name, location or category so that I can quickly find places that interest me.	Implement the approved search bar. Create the search query/controller logic. Match keyword against attraction name, location and category. Update the existing integrated attraction list. Handle blank keywords and repeated searches. Test partial, case-insensitive and no-match searches.	High	5
PB09	As a guest or registered user, I want to filter attractions by category, location area and minimum rating so that I can refine the attraction list.	Implement Category, Location Area and Minimum Rating controls. Create filter query/controller logic. Allow filters to work without a keyword. Update the integrated attraction list. Handle invalid or unavailable filter values. Test each filter individually.	High	5
PB10	As a guest or registered user, I want to combine a search keyword with selected filters so that I can narrow the results more precisely.	Combine the search and filter query conditions. Preserve selected filters when the keyword changes. Preserve the keyword when filters change. Refresh the same result list after every change. Handle combinations that return no results. Test common and edge-case combinations.	High	3

PB11	As a guest or registered user, I want to see the current result count and applied filters so that I understand why particular attractions are displayed.	Display the number of matching attractions. Show active keyword and selected filters in the approved UI. Update the count and indicators after every query change. Hide inactive filter indicators. Support removal of an individual applied filter. Test the indicators and count.	Medium	2
PB12	As a guest or registered user, I want to clear my keyword or selected filters so that I can restore the default Melaka attraction list.	Add clear-keyword and clear-filter actions. Add a clear-all/reset action. Reset the corresponding query state. Restore the default Melaka attraction list. Reset result count and applied-filter indicators. Test individual and full reset flows.	Medium	2
PB13	As a guest or registered user, I want to receive a helpful no-result state so that I can change my keyword, clear filters or return to the default list.	Implement the approved no-result state on the same page. Display the active keyword and filters causing no match. Provide change-keyword, clear-filter and reset-list actions. Keep search and filter controls usable. Avoid navigation to a separate error page. Test no-result recovery.	Medium	2
PB14	As a guest or registered user, I want to view an attraction's name, description, category, address and rating so that I can learn about it before visiting.	Implement the approved Attraction Details page. Create attraction-details retrieval logic. Display name, description, category, address and rating. Link attraction cards to the correct details record. Add loading and attraction-not-found states. Test correct data mapping.	High	5
PB15	As a guest or registered user, I want to view photo previews for an attraction so that I can see what the location looks like.	Implement the approved photo-preview area. Retrieve the selected attraction's photo URLs. Display responsive photo thumbnails/previews. Load images without blocking the remaining details page. Handle individual image-load failure. Test one-photo and multi-photo attractions.	Medium	3

PB16	As a guest or registered user, I want to enlarge a selected attraction photo and use previous, next and close controls so that I can inspect the photos comfortably.	Implement the optional photo lightbox. Open the selected preview at the correct image. Add previous, next and close controls. Handle first/last image navigation consistently. Support keyboard or accessible close behaviour where feasible. Test all controls.	Medium	3
PB17	As a guest or registered user, I want the attraction details to remain usable when no photo is available so that missing images do not block the page.	Implement the approved no-photo fallback. Detect an empty or unusable photo list. Display non-blocking fallback content. Keep all non-photo details and map entry accessible. Avoid opening an empty lightbox. Test missing and broken photo scenarios.	Medium	2
PB18	As a guest or registered user, I want to view one selected attraction on a map with its name, address, marker and zoom or pan controls so that I can inspect its location.	Implement the approved map section for one attraction. Load the selected attraction's coordinates. Display one marker, name and address. Enable basic zoom and pan controls. Exclude current location, directions, route, travel time and transport mode. Test marker accuracy and controls.	Medium	3
PB19	As a guest or registered user, I want to see the attraction address in text form when the map is unavailable so that I can still obtain its location information.	Detect map-loading or map-service failure. Display the approved map-unavailable state. Keep the attraction name and address visible in text. Avoid broken map controls or blank containers. Allow retry when appropriate. Test unavailable and timeout cases.	Medium	2
PB20	As a guest or registered user, I want previously loaded attraction lists, details, images, searches and map regions to remain viewable during a weak or unavailable connection.	Define the cacheable Attraction Explorer resources. Cache the application shell and previously loaded attraction data. Cache eligible images and map-region resources within service limits. Show an offline/weak-connection message. Use cached data without presenting stale data as live. Test online-to-offline behaviour.	Medium	5

PB21	As a registered user, I want to create a review for an attraction so that I can share my experience with other travellers.	Implement the approved review form. Create the review model and persistence logic. Associate the review with the logged-in user and attraction. Validate required review text. Save and display the new review. Test authorized and unauthorized submission.	High	5
PB22	As a registered user, I want to give a rating from one to five stars when submitting a review so that I can evaluate the attraction.	Implement the approved 1–5 star input. Require one valid rating during review submission. Validate the numeric rating range. Store the rating with the review. Display the submitted rating. Test each valid value and invalid input.	High	2
PB23	As a registered user, I want to upload photos together with my review so that other travellers can see my actual experience.	Implement the approved photo-selection and preview UI. Validate supported image type and size. Upload selected images to image storage. Save uploaded image URLs with the review. Display uploaded photos with the review. Handle partial or failed uploads and test.	High	3
PB24	As a registered user, I want to edit my own review so that I can correct or update it later.	Show the edit action only for the review owner. Load the existing review into the approved edit form. Validate updated text, rating and photos. Save permitted changes. Refresh the displayed review. Test ownership and update scenarios.	Medium	3
PB25	As a registered user, I want to delete my own review so that I can remove content that I no longer want to share.	Show the delete action only for the review owner. Display the approved confirmation dialog. Delete the selected review and linked references safely. Refresh the attraction's review list. Update dependent rating/map data where required. Test cancel, confirm and unauthorized deletion.	Medium	2

PB26	As a guest or registered user, I want to view reviews posted under an attraction so that I can learn from other travellers.	Retrieve reviews for the selected attraction. Display reviewer profile information, rating, text and photos. Order reviews using the confirmed default order. Add loading and no-review states. Link reviewer information to the correct profile where permitted. Test data isolation by attraction.	High	3
PB27	As a registered user, I want clear validation messages for incomplete review text, missing ratings or invalid photos so that I can correct my submission.	Define validation rules for review text, rating and photos. Display field-level messages in the approved UI. Preserve valid entered content after validation failure. Prevent invalid review persistence. Handle upload and network validation errors. Test all required validation cases.	Medium	2
PB28	As a user, I want the attraction's average rating to update when valid ratings are submitted so that the displayed rating reflects the available reviews.	Calculate average rating from valid review ratings. Recalculate after create, edit or delete review operations. Store or derive the value consistently. Display the updated average on attraction list/details. Handle an attraction with no ratings. Test rounding and synchronization.	High	3
PB29	As a registered user, I want to view my personal exploration map so that I can visualise my travel activity in Melaka.	Implement the approved personal Exploration Map page. Retrieve supported Melaka attraction coordinates. Display the base map and supported markers. Load data for the logged-in user. Add loading and map-unavailable states. Test marker accuracy and user isolation.	High	5
PB30	As a registered user, I want attractions that I have reviewed to be highlighted as visited so that my map represents the places I have explored.	Derive visited attractions from the user's submitted reviews. Retrieve the user's reviewed attraction IDs. Apply the approved visited-marker style. Keep unvisited markers visually different. Refresh visited status after relevant review changes. Test multiple users and review changes.	High	5

PB31	As a registered user, I want to view a list of my visited attractions so that I can review the places I have explored.	Implement the approved visited-locations list. Retrieve the user's visited attractions from review records. Display attraction name and confirmed summary information. Link list items to the relevant attraction details. Add empty and loading states. Test list/map consistency.	Medium	3
PB32	As a registered user, I want to view my exploration progress so that I know how many supported attractions I have completed.	Define the supported-attraction total used for progress. Count the user's distinct visited attractions. Calculate the progress value and percentage. Display the approved progress summary. Update progress after review changes. Test zero, partial and complete progress.	Medium	3
PB33	As a registered user, I want my exploration map, visited list and progress to update after I submit a review so that all travel records remain consistent.	Connect review create/edit/delete events to exploration records. Refresh the visited marker state. Refresh the visited-locations list. Recalculate progress. Prevent duplicate visited records for multiple reviews of one attraction. Test cross-module synchronization.	High	3
PB34	As a registered user, I want clear empty and map-unavailable states so that the Exploration Map remains understandable when I have no visits or the map cannot load.	Implement the approved no-visited-attractions state. Implement map-loading and map-unavailable states. Keep visited count and explanatory text visible where possible. Provide a safe retry action where appropriate. Avoid displaying misleading progress. Test no-data and service-failure cases.	Medium	2
PB35	As a registered user, I want to view another user's public profile so that I can learn about their travel activities.	Implement the approved public-profile page. Retrieve the selected user's public profile data. Display name, profile picture and confirmed public summary. Prevent private fields from being exposed. Add loading and user-not-found states. Test selected-user accuracy.	Medium	3

PB36	As a registered user, I want to view reviews written by another user from their profile so that I can learn from their experiences.	Retrieve reviews created by the selected user. Display the reviews within or from the public profile. Show related attraction, rating, text and photos where available. Link to the relevant attraction details. Add no-review state. Test selected-user filtering.	Medium	3
PB37	As a registered user, I want to view another user's exploration map so that I can see the attractions they have visited.	Retrieve the selected user's visited attraction data. Display the approved public exploration map. Highlight only that user's visited attractions. Keep private controls unavailable. Add no-data and map-unavailable states. Test selected-user isolation.	Medium	5
PB38	As a registered user, I want to compare my exploration progress with another user so that I can understand the difference between our progress.	Retrieve progress summaries for the current and selected user. Display both values using the approved comparison UI. Calculate the numeric difference. Label each user's data clearly. Handle missing progress data. Test comparison accuracy.	Medium	5
PB39	As a registered user, I want to compare my visited attractions with another user so that I can identify common and different visited places.	Retrieve both users' distinct visited-attraction sets. Identify common attractions. Identify attractions unique to each user. Display results using the approved comparison UI. Link attractions to details where permitted. Test empty, identical and different sets.	Medium	5
PB40	As a registered user, I want clear user-not-found and no-data states so that I understand when another user has no public reviews or exploration records.	Implement user-not-found state. Implement no-review, no-map and no-progress states. Keep available profile data visible when only one section lacks data. Avoid exposing internal errors or private data. Provide safe navigation back to the previous page. Test each missing-data combination.	Medium	2

PB41	As a registered user, I want to add and remove attractions from my wishlist so that I can remember places that I may visit in the future.	Implement the approved wishlist action on attraction UI. Create wishlist persistence for each user. Add and remove selected attractions. Prevent duplicate wishlist entries. Display the approved Wishlist page and empty state. Test user isolation and Guests.	Medium	3
PB42	As a registered user, I want to add and remove favourite attractions so that I can easily access the places that I like most.	Implement the approved favourite action. Create favourite persistence for each user. Add and remove selected attractions. Prevent duplicate favourite entries. Display the approved Favourite Attractions page and empty state. Test user isolation and Guests.	Medium	3
PB43	As a registered user, I want to view all reviews that I have created so that I can manage and revisit my previous reviews.	Implement the approved My Reviews page. Retrieve reviews created by the logged-in user. Display related attraction, rating, content and date. Link to edit/delete flows and attraction details. Add loading and empty states. Test account isolation.	Medium	3
PB44	As a registered user, I want to view all photos that I have uploaded through my reviews so that I can revisit my shared travel photos.	Implement the approved My Uploaded Photos page. Retrieve photos from the logged-in user's reviews. Display thumbnails grouped or linked to the related review/attraction. Exclude deleted or unavailable photos. Add loading and empty states. Test account isolation.	Medium	3
PB45	As a registered user, I want to view my personal travel history in chronological order so that I can revisit my previous travel activities.	Define the travel-history event source based on confirmed reviewed/visited records. Retrieve the logged-in user's historical records. Order records chronologically. Display attraction and confirmed activity summary. Link to related attraction/review where available. Add loading and empty states and test.	Medium	5